

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Regular/Supplementary Summer Examination – July 2025****Course: B.Tech****Branch: Computer Engineering /Allied****Semester: VII****Subject Code & Name: BTCOE705B (Deep Learning)****Max Marks: 60****Date :25/07/2025****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Which of the following is/are Common uses of RNNs?		1				
	<table> <tr> <td>a. Generate analytic reports</td> <td>b. Detect fraudulent credit-card transaction</td> <td>c. Provide a caption for images</td> <td>d. All of these</td> </tr> </table>	a. Generate analytic reports	b. Detect fraudulent credit-card transaction	c. Provide a caption for images	d. All of these		
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2	. Which of the following is well suited for perceptual tasks?		1				
	<table> <tr> <td>a. Feed-forward neural networks</td> <td>b. Recurrent neural networks</td> <td>c. Convolutional neural networks</td> <td>d. Reinforce - ment Learning</td> </tr> </table>	a. Feed-forward neural networks	b. Recurrent neural networks	c. Convolutional neural networks	d. Reinforce - ment Learning		
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3	CNN is mostly used when there is an		1				
	<table> <tr> <td>a. structured data</td> <td>b. unstructured data</td> <td>c. Both a and b</td> <td>d. None of these</td> </tr> </table>	a. structured data	b. unstructured data	c. Both a and b	d. None of these		
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4	Which of the following is/are Limitations of deep learning?		1				
	<table> <tr> <td>a. Data labelling</td> <td>b. Obtain huge training datasets</td> <td>c. Both a and b</td> <td>d. None of these</td> </tr> </table>	a. Data labelling	b. Obtain huge training datasets	c. Both a and b	d. None of these		
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5	In which of the following applications can we use deep learning to		1				

	solve the problem?		
	a. Protein structure prediction	b. Prediction of chemical reactions	c. Detection of exotic particles
			d. All of these
6	Sentiment analysis using Deep Learning is a many-to one prediction task-		1
	a. True	b. False	c.
			d.
7	In CNN, having max pooling always decrease the parameters?		1
	a. True	b. False	c.
			d.
8	Which of the following is a common loss function used for binary classification in deep learning?		1
	a. Mean Absolute Error (MAE)	b. Mean Squared Error (MSE)	c. Binary Cross-Entropy
			d. Categorical Cross-Entropy
9	Which activation function is commonly used in the hidden layers of a deep neural network?		1
	a. ReLU	b. Sigmoid	c. softmax
			d.tanh
10	Which of the following is a common optimization algorithm used in deep learning?		1
	a. Gradient Descent	b. Stochastic Gradient Descent	c. Adam
			d.All of these
11	Which of the following is a common activation function used in the output layer for binary classification?		1
	a. ReLU	b. Sigmoid	c. softmax
			d.tanh
12	Which deep learning technique is used for unsupervised feature learning?		1
	a. Autoencoders	b. CNN	c. RNN
			d.Reinforce - ment Learning
Q. 2	Solve the following.		12
A)	State and explain key differences between machine learning and		6

	deep learning.		
B)	State and explain applications of deep learning.		6
Q.3	Solve the following.		12
A)	Explain the architecture of feed forward neural network with necessary conventions.		6
B)	What is back propagation learning? How back propagation learning algorithm is used to update weight vectors?		6
Q. 4	Solve Any Two of the following.		12
A)	Explain architecture of multilayer perceptron.		6
B)	Define and explain following terms. (i) Cross Entropy loss (ii) Local Gradient (iii) Back propagated Gradient		6
C)	What is Autoencoder? Explain sparse and undercomplete autoencoder in detail.		6
Q.5	Solve Any Two of the following.		12
A)	Which are the different layers of Convolutional Neural Network? Explain their functions.		12
B)	What is vanishing gradient problem? What are the challenges in Gradient Descent?		12
C)	What is overfitting and underfitting problem? How is it resolved in neural networks?		
Q. 6	Solve Any Two of the following.		12
A)	What is the need of normalization? Explain different kinds of normalization techniques in neural networks.		6
B)	How face recognition process is carried using Deep Learning approach?		6
C)	Explain Recurrent Neural network.		6
	*** End ***		